

SAI ROHITH BHARADWAJ RAMAVARAPU

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EDUCATION

University at Buffalo, SUNY — M.S. Computer Science, Artificial Intelligence
VNR Vignana Jyothi Institute of Engineering & Technology — B.Tech, Computer Science

Expected Dec 2026

May 2024

TECHNICAL SKILLS

Languages: Python, C#, Java, C++, SQL, TypeScript, JavaScript

Backend: ASP.NET Core, FastAPI, REST APIs, Microservices, JWT, PostgreSQL, MySQL, Redis

Cloud & Tools: AWS, Docker, CI/CD, Git, Linux, performance profiling

AI / ML: PyTorch, OpenCV, Scikit-learn, Hugging Face, vector search

EXPERIENCE

University at Buffalo — Graduate Research Assistant, Robotics & Vision

Aug 2026 – Present

- Developing adaptive illumination software to improve Visual SLAM robustness in extreme low-light environments.
- Built PyTorch training and evaluation workflows for illumination decomposition and behavior-cloned control policies.

UST — Software Engineer

Dec 2024 – Jun 2025

- Built and deployed 12+ C# / ASP.NET Core REST microservices for carbon-emissions analytics across transportation, waste, and supply chains.
- Reduced API response latency by approximately 30% through SQL indexing, connection pooling, complexity improvements, and asynchronous data access.
- Implemented JWT authentication, role-based access control, and structured logging for secure, diagnosable services.
- Delivered Angular components with reactive forms and client-side validation for emissions-data workflows.

Tech Mahindra — Software Engineer Trainee

Sep 2023 – Nov 2023

- Built a PDF question-answering pipeline using sentence embeddings, semantic chunking, cosine similarity, and vector search; recorded 0.89 F1 on project evaluation.

South Matrix Institute of Technology — Machine Learning Intern

Jun 2024 – Dec 2024

- Processed and analyzed 500K+ transaction records with Python and Pandas and engineered features for downstream models.

PROJECTS

ShadowFix: Real-World Shadow Removal / *Python, PyTorch, OpenCV, Gradio*

- Engineered a reproducible evaluation pipeline comparing RGB, residual, and mask-guided U-Nets on 540 held-out ISTD triplets.
- Analyzed a best result of 30.189 dB PSNR and 0.9514 SSIM, including +3.01 dB overall and +2.21 dB shadow-region PSNR over the RGB baseline.
- Validated automatic and painted-mask Gradio workflows and documented failure cases to guide reliable real-world inference.

Video Anomaly Detection / *Python, PyTorch, OpenCV, CNN-LSTM*

- Implemented a modular video-processing pipeline combining frame sampling, motion cues, CNN spatial features, and LSTM sequence modeling for anomaly scoring.
- Separated preprocessing, training, and inference into reusable stages supporting both recorded inputs and live video streams.

Clinical Risk Prediction API / *FastAPI, Scikit-learn, React, Docker*

- Benchmarked Random Forest, SVM, and gradient-boosted classifiers with standardized preprocessing and validation, achieving 0.92 ROC-AUC.
- Designed containerized REST endpoints for validated model inference and connected them to a responsive React user workflow.

PUBLICATION

“Smart Health Prediction System with Data Mining.” Co-author, *Nanotechnology Perceptions*, Vol. 20, S15, 2024. [Article](#)